RescueSwarm: An Indigenously Sourced Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Research and Development Funding Proposal

Swastik Kumar
Department of Electronics & Communication Engineering
Thapar Institute of Engineering & Technology
Patiala, Punjab, India
skumar6_be24@thapar.edu

***Abstract*—India recorded 5.4 million internal displacements from disasters in 2024, the highest figure in twelve years, and the binding constraint on survivor outcomes is the speed with which casualties are located and reached. Ground and boat-based search is slow and hazardous; helicopter search is fast but scarce and expensive. This proposal describes RescueSwarm, a three-aircraft autonomous unmanned aerial system that surveys a designated area, detects and geolocates survivors, and delivers a standardised relief kit to each, under a single operator and without any reliance on cellular or internet infrastructure. The system is designed around a deliberate constraint: indigenous sourcing wherever the budget permits it, both because supply-chain sovereignty is a national priority and because domestic lead times are roughly half those of imports, which directly buys flight-test days inside a schedule-limited programme. The configuration costed here reaches approximately 36 % Indian content, and I state which lines were deferred to reach the requested figure and what restoring them would cost. I present the system architecture, the sizing and error models that fix its design point, the simulation evidence obtained to date, a market-verified cost basis, and a phased work plan. I am explicit throughout about which figures are measured, which are modelled, and which remain assumed; no aircraft has yet flown, and the proposal is structured so that a reviewer can distinguish demonstrated capability from projected capability without reading the source material. The requested funding is INR 7.65 L over an eight-month build.**

***Index Terms*—unmanned aerial systems, multi-robot coordination, search and rescue, coverage path planning, RTK geolocation, edge inference, indigenous manufacturing, disaster response**

I. Introduction and Problem Statement

A. The operational gap

India records among the highest disaster-displacement figures in the world. The Internal Displacement Monitoring Centre reports **5.4 million internal displacements in India during 2024** — the highest in twelve years — of which 2.4 million were triggered by a single monsoon flood season in Assam [8]. The scale of the response problem is therefore not in question; what is in question is how quickly the people displaced by it can be found.

Post-flood search and rescue in India is dominated by a single quantity: the elapsed time between the onset of the event and the moment a survivor is both *located* and *reached*. Nationally deployed capability addresses the second of these far better than the first. Boat teams and ground parties can reach a survivor once told where to go, but they search slowly, at high risk to the crew, and with poor coverage of debris fields and rooftops. Rotary-wing aircraft search quickly but are scarce, costly per flight hour, and are typically committed to evacuation rather than to systematic area survey.

The consequence is that *localisation* is the bottleneck. A search asset that can systematically cover an assigned polygon, detect human figures in cluttered post-flood imagery, and report each detection as a coordinate accurate to a few metres would materially change the tasking of every downstream resource.

B. Why a swarm rather than a single aircraft

A single multirotor with a useful payload and a fifteen-minute endurance covers a disappointing area. Coverage scales with the number of simultaneously operating aircraft, but so does operator workload, and a system requiring one pilot per aircraft does not deploy well in a disaster. The engineering problem is therefore not simply flight, but *coordinated autonomous flight under a single operator*: area decomposition, task allocation, deconfliction, consolidated reporting, and graceful degradation when the command link is lost.

C. Why indigenous sourcing is a design constraint, not a slogan

I treat domestic sourcing as a first-class design constraint for two reasons, only one of which is political.

The first is sovereignty: a disaster-response asset whose critical components have a single foreign supplier is a strategic liability, and national policy correctly prioritises domestic capability.

The second is schedule, and in a programme of this length it dominates. My supplier survey puts typical import lead times at four to six weeks against two to three weeks domestically. In a programme whose binding constraint is an eight-week flight-test window, that difference is not a procurement detail

— it is the difference between two test campaigns and one. Indigenisation, in this programme, buys flying days.

That argument cuts against a decision taken later in this document, and I would rather confront it than let a reviewer find it. Section VII-D defers the Indian autopilot and Indian propulsion on cost, and the imported parts replacing them carry the longer lead time this section has just described as decisive. The mitigation is structural rather than rhetorical: every long-lead item is ordered in tranche 1, in month one, before the software work that would normally precede procurement is finished. Cost was traded against schedule margin knowingly, and the schedule risk it creates is carried explicitly in Table IX rather than assumed away.

D. *Contributions of the proposed work*

1) A three-aircraft autonomous survey-and-delivery system operable by a single operator with no network dependency, sized and costed line-by-line against market-verified components with dated Indian supplier prices.
2) A geolocation error budget that treats survivor coordinate accuracy as the primary system output, and an architecture that meets it using a team-owned RTK base rather than any external correction service.
3) A coordination scheme — static equal-area decomposition with boustrophedon coverage and greedy claim-and-lock task allocation — selected for determinism and verifiability over optimality.
4) An evidence discipline in which every published number regenerates from its model in continuous integration, and in which measured, modelled and assumed quantities are separated by construction.

II. Objectives and Scope

A. *Primary objectives*

O1 Survey a designated area autonomously with three cooperating aircraft under one operator and one ground control station.

O2 Detect human figures in nadir imagery at a ground sample distance sufficient for reliable classification, with onboard inference and multi-frame temporal fusion.

O3 Report each detection as a geodetic coordinate with a circular error probable within a few metres.

O4 Deliver a standardised relief kit to each confirmed survivor location by controlled free-fall release.

O5 Operate with no cellular, internet or cloud dependency at any point in the mission.

O6 Maximise value-weighted indigenous content across the priced bill of materials subject to the funding available, without compromising O1–O5, and state explicitly which Indian lines were deferred to meet the budget and what restoring them costs.

B. *Explicit non-objectives*

I state these because proposals of this kind are frequently over-scoped, and because each exclusion is a deliberate engineering decision rather than an omission.

- **Not a casualty-evacuation platform.** The payload class is a relief kit, not a person, and no growth path to the latter is claimed.
- **Not beyond-visual-line-of-sight certified.** The design operates within a bounded geofence; regulatory expansion is future work.
- **Not a night or all-weather system.** The perception stack is visible-band. Thermal imaging is identified as the highest-value future extension but is outside this programme.
- **Not a survivor-triage system.** The system reports position and delivers a kit; it makes no medical assessment.

III. Related Work and State of the Art

A. *Coverage path planning*

Area decomposition for multi-robot coverage is mature. Divide Areas for Optimal Multi-Robot Coverage (DARP) produces equal-area, connected, robot-specific regions and is the natural reference method [1]. Classical boustrophedon decomposition [2] remains the standard in-region motion primitive for sensor footprints that are rectangular in the body frame.

I adopt a static equal-area partition with boustrophedon in-region coverage rather than a full DARP implementation. This is a deliberate trade of optimality for determinism: a static partition produces an identical, inspectable plan for identical inputs, which is a precondition for the verification regime described in Section V-A. DARP is retained as design rationale and as a future-work path where dynamic reallocation becomes necessary.

B. *Multi-robot task allocation*

The Consensus-Based Bundle Algorithm (CBBA) is the canonical decentralised auction method for multi-agent assignment [3]. Its guarantees depend on a consensus phase over a communication graph. For a three-aircraft system with a reliable star topology to a single ground station, I implement greedy claim-and-lock with a deterministic tie-break, which is materially simpler to verify and whose worst-case behaviour on three agents is well within acceptable bounds.

C. *Aerial detection of people*

Small-object detection from altitude is the governing difficulty: a supine adult at survey altitude occupies a few tens of pixels. Slicing-aided hyper inference (SAHI) and related tiling strategies substantially improve small-object recall by running detection on overlapping crops rather than a downsampled full frame [4]. Purpose-built aerial person datasets such as SARD [5] and the broader VisDrone benchmark [6] provide the transfer-learning base; neither contains post-flood Indian terrain. A field data campaign is therefore the right answer and was costed; it is *deferred* in the present ask (Section VII-E), which is why detection recall remains a modelled quantity and the leading entry in the risk register.

TABLE I
AIR VEHICLE DESIGN POINT (PER AIRCRAFT)

Parameter	**Value**
Configuration	Quadrotor
Rotor diameter	18 in
Maximum take-off mass	6.36 kg
Fleet mass (3 aircraft)	19.08 kg
Regulatory fleet cap	25 kg
Battery	6S3P Li-ion 21700, 292 Wh
Cells per aircraft	18
Static thrust-to-weight	2.0
Hover endurance	15.3 min
Design mission duration	7.7 min
Survey altitude	60 m AGL
Ground sample distance	≤2.0 cm/px
Survey groundspeed	8 m/s
Payload	4 kits × 200 g
Geofence radius	600 m

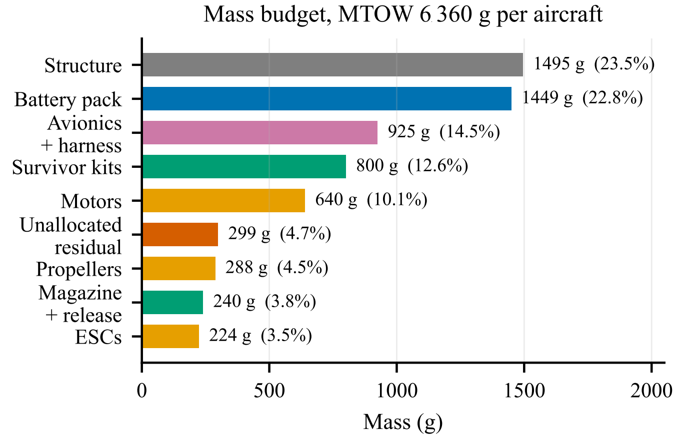

Fig. 1. Mass budget per aircraft. Generated from `docs/sizing/model-output.txt`. The fleet figure of 19 081 g carries 4 919 g of growth allowance against a 24 kg internal target — 26 % build overweight tolerated — inside the 25 kg regulatory cap.

D. Airborne payload delivery

The dominant error term in airborne delivery is release state — velocity and wind — rather than release altitude. Mathisen *et al.* demonstrate autonomous ballistic airdrop from a small fixed-wing UAV and report a mean delivery error of 5.5 m, with the release state computed dynamically against wind and vehicle velocity and re-optimised in flight precisely because that term dominates [7]. This is the decisive argument for a multirotor in this role: a rotorcraft can null its groundspeed to near zero before release and thereby remove the dominant error term outright, which a fixed-wing cannot. The delivery budget in Section IV-G exploits exactly this.

E. Positioning

Real-time kinematic correction from a locally surveyed base is standard practice and is the only route to sub-metre relative positioning without an external network [11]. Since the system may not use internet or cellular services, a team-owned base station is not merely preferable but required.

IV. PROPOSED SYSTEM ARCHITECTURE

A. Design point

Table I states the aircraft design point. Every value is produced by a sizing model held in the repository and regenerated in continuous integration; none is transcribed by hand.

This is the reference design point, and one line of it is not yet verified for the aircraft actually being funded. The adopted configuration (Section VII-D) uses generic motors that publish no thrust curve, so the thrust-to-weight figure below is a requirement rather than a measurement until the P5 thrust stand runs. Nothing else in the table moves: mass, energy, altitude, speed and payload are all set by the mission.

The thrust-to-weight ratio of 2.0 and the hover endurance of 15.3 minutes against a 7.7-minute design mission are both reserve policies rather than performance targets: the mission is sized to be completed comfortably within half the available energy, because the failure mode that ends a competition or a deployment is an aircraft that does not return.

Fig. 1 gives the mass statement behind those figures. Two features drive the rest of the design. Structure and battery together are 46 % of maximum take-off mass, which is why airframe mass fraction and pack chemistry are the two sensitivities carried in the risk register rather than, say, avionics choice.

The itemisation does not close, and the figure shows the gap rather than hiding it. The model's listed items sum to 6 061 g against a 6 360 g maximum take-off mass, leaving 299 g — 4.7 % — unallocated. That residual is small enough not to threaten the fleet margin and large enough that it should be attributed rather than absorbed silently; closing it is a P1 action. The survivor kits are 800 g — 12.6 % of MTOW and fixed by the mission at four kits of exactly 200 g — so payload is an input to the sizing loop, not an output of it. This is why propulsion and energy resist cost reduction in Section VII in a way that avionics does not: they are sized by a payload that is not negotiable, so cheaper parts of the same class are available but a smaller aircraft is not.

B. Segment architecture

The system comprises three segments, shown in Fig. 2.

1) Air segment: Three identical quadrotors, each carrying a Pixhawk-standard autopilot running ArduPilot [10], a dual-band NavIC/GNSS receiver providing RTK position, a nadir-mounted 12 MP camera, an edge inference module, a four-station kit magazine with positive-lock detents, a 5.8 GHz mesh radio for telemetry and video, and a 2.4 GHz ExpressLRS link carrying safety-pilot control and abort independently of the autonomy stack.

2) Ground segment: A single ground control station presenting one unified operator interface. The station is architecturally incapable of originating a retask, waypoint change or drop command; abort and recall are the only permitted

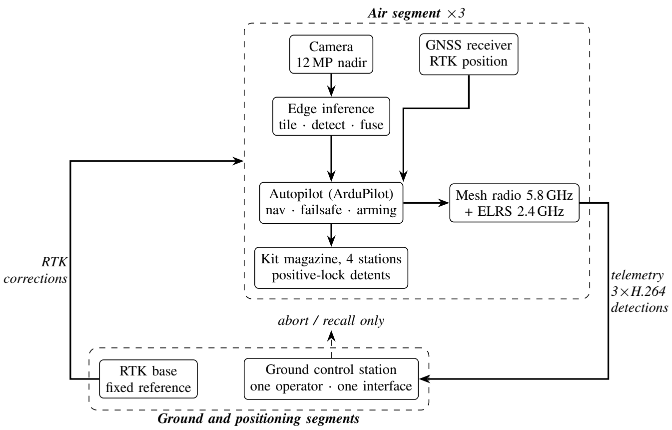

Fig. 2. System architecture. The only permitted downlink commands are abort and recall; the ground station is architecturally incapable of originating a retask, waypoint change or drop command. Detection runs onboard, so the downlink serves the operator rather than the algorithm.

downlink commands, and are implemented as an explicit exception. This is a structural safety property rather than a policy: the relevant code paths are absent from the deployed build, not merely guarded.

3) Positioning segment: A team-owned RTK base station, positioned and set to a fixed reference from its first three-dimensional fix within the setup window. Launch is gated on a three-dimensional fix rather than an RTK fix; the first geotag is gated on RTK-fixed, and detections made before convergence are geotagged in float and re-fused once fixed. This decoupling recovers approximately ninety seconds of setup time that a survey-in procedure would otherwise consume.

C. Geolocation error budget

Survivor coordinate accuracy is the primary system output, and the error budget is therefore the governing analysis of the whole design. Total error is the root sum of squares of GNSS position error, boresight and lever-arm residual, attitude error, terrain-model error and timing skew.

Table II shows the dominant conclusion: positioning quality, not optics, governs. This is what makes RTK a scoring requirement rather than an optimisation, and it is why the RTK receiver and its base station survive every cost-reduction pass in Section VII while a great deal else does not. The *second* receiver, which supplies moving-baseline heading, is a different case and is deferred to a measurement — see Section VII.

Multi-frame fusion is effectively free. At 60 m altitude and 8 m/s groundspeed, forward overlap yields on the order of fourteen independent observations of each target per pass even at a conservative 2 Hz inference rate. Those observations are used both for temporal confirmation, which suppresses false positives, and for geolocation averaging.

Fig. 3 shows both effects quantitatively. Panel (a) makes the positioning argument unarguable: losing the RTK fix moves the error by nearly an order of magnitude, and a 3-D-only fix at 3.67 m RSS consumes most of the 5 m delivery allowance

TABLE II
GEOLOCATION ERROR BY CONFIGURATION

Case	Configuration	RSS error
A	No RTK, single-frame	3.09 m
B	RTK, single-frame	∼1.3 m
C	RTK + multi-frame fusion + calibrated ground plane	**0.91 m**

Case C assumes moving-baseline heading. That receiver is deferred (Section VII-E), which substitutes a magnetometer-class attitude term and gives **0.94 m** — an 0.03 m penalty, and the reason the deferral is judged affordable.

before ballistics or position-hold error are counted. Panel (b) shows why fusion is worth having but not worth over-investing in — it divides the random terms by $\sqrt{n}$ and leaves boresight and target extent untouched, so the curve departs from the ideal $1/\sqrt{n}$ line and flattens by about five frames. Twenty frames is not twice as good as five, and the roughly fourteen frames the geometry supplies for free sit comfortably past the knee.

D. Perception and inference budget

A 12 MP sensor at 60 m yields a ground sample distance of approximately 2 cm/px, at which a supine adult of 1.7 m subtends about 85 pixels. Running detection on that full-resolution frame would require 48 tiles per frame and is far beyond the thermal and compute envelope of any module in this mass class.

The design therefore downsamples by two before tiling. That coarsens the ground sample distance to approximately 3.65 cm/px and reduces the same target to **about 47 pixels** — small, but within reach of tiled inference — while cutting the tile count to 12 per frame at 20 % overlap. At a 2 Hz inference rate this is 24 inferences per second at 640×640, which is within the demonstrated throughput of the 26-TOPS-class accelerator selected in Section VII. The 47-pixel figure, not the 85-pixel one, is what the detector actually sees, and it is the number the recall risk is written against.

This is the single most benchmark-sensitive figure in the design. Published throughput for comparable models on comparable hardware spans roughly 24 to 41 frames per second, and the lower end of that range leaves no margin. Hardware benchmarking before design freeze is accordingly a funded, scheduled activity rather than an assumption.

E. Onboard processing pipeline

Fig. 4 shows what runs where. The division of labour is the part worth stating plainly, because it is what keeps the search phase deterministic: **the autopilot owns the flight, and the companion computer never touches it during the sweep.** Transects are uploaded as an ordinary AUTO mission during setup, and the aircraft would fly the entire search pattern with the companion powered off. The companion observes, detects, and takes control only for a delivery excursion — a bounded GUIDED interruption that returns the aircraft to the same AUTO mission afterwards.

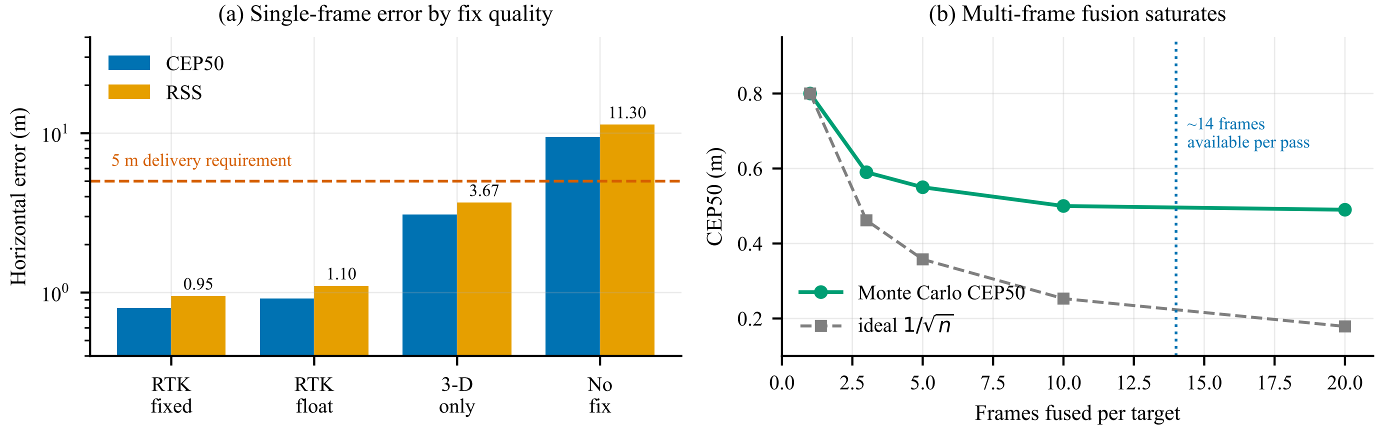

Fig. 3. Geolocation error, from a Monte Carlo run through the same projection code the aircraft flies. (a) Single-frame horizontal error against GNSS fix quality, typical attitude, on a logarithmic scale; the dashed line is the 5 m delivery requirement. (b) Effect of fusing multiple observations of the same target, RTK fixed. Generated by `figures/make_figures.py` from `docs/sizing/geotag-accuracy-output.txt`.

That boundary matters for two reasons. A companion crash or thermal throttle costs detections, not control. And the search path is reproducible from the mission file alone, which is what makes the coverage and deconfliction results in Section V-A checkable rather than emergent.

Detection does not gate flight. Inference runs on captured frames while the aircraft continues its transect. Losing frames to a thermal event or a slow model costs looks at a target, and the fusion requirement of twelve looks per pass carries enough redundancy to absorb some of that; it does not stall the sweep or extend the mission.

The accelerator is a substitution the inference budget has not yet been re-run against. The tiling arithmetic above was derived against a Jetson-class module, whose published throughput for comparable models spans roughly 24 to 41 frames per second. The adopted configuration (Section VII) instead pairs a 26-TOPS accelerator with a single-board host, which is a different architecture with a different throughput curve and a lower sustained power draw. The 24-inferences-per-second requirement is therefore stated against hardware that has not been benchmarked, and P5 benchmarking on the parts actually purchased is the gate before design freeze. Table IX carries this rather than the text implying the figure is settled.

F. Communications

Regulatory rules require the ground station to display a live camera feed from *each* aircraft. Three independent MJPEG streams would demand 4.5–6 Mbps against a link budget of approximately 2.5 Mbps. The system therefore uses H.264 encoding served through a media gateway, carrying three concurrent WebRTC streams in approximately 2.7 Mbps, with per-feed rate reduction under congestion rather than feed switching. Detection runs onboard; the downlink serves the operator, not the algorithm.

Band diversity is provided without a third radio. The 5.8 GHz mesh carries video, telemetry and coordination; a 2.4 GHz ExpressLRS link carries safety-pilot control and abort, and is independent of the autonomy stack. Both operate in delicensed bands and both comfortably cover the 600 m geofence.

The abort path is not an assumption about the link; it is a documented mode of it. ExpressLRS carries MAVLink natively from version 3.5, so the same receiver that gives the safety pilot manual control also carries the autopilot's command and telemetry stream — the ArduPilot side is a serial port set to MAVLink v2, nothing more [9]. The earlier AirPort mode achieves the same thing as a transparent serial bridge and is no longer the recommended route for MAVLink. Naming the mechanism matters here: deferring the sub-gigahertz radio (Section VII-E) rests entirely on this link being able to do both jobs. A further link in the 865–867 MHz short-range-device band would add a third independent path and is the correct extension when funding allows; it is deferred rather than fitted, and Section VII-E records the reasoning.

G. Payload delivery

Kits are released at 6 m above ground level, gated on a residual groundspeed below 0.3 m/s. At that release state, ballistic dispersion contributes approximately 0.32 m to total delivery error; combined with geolocation and position-hold error the root-sum-square is approximately 1.5 m with RTK active. Release altitude is chosen low enough that wind drift remains under a metre in realistic conditions, and high enough for safe clearance above a person and above debris.

Positive-lock mechanical detents on each station ensure that a power interruption cannot release a kit. This is treated as a safety-critical item and is excluded from cost reduction by policy.

H. Mission sequence

Fig. 5 shows the autonomous mission sequence. Two properties are worth drawing out. First, the launch stagger lives in the mission file as an explicit delay before each take-off command, not in operator timing — this is what converts a 1.31 m closest approach into the 64.80 m of Fig. 6. Second,

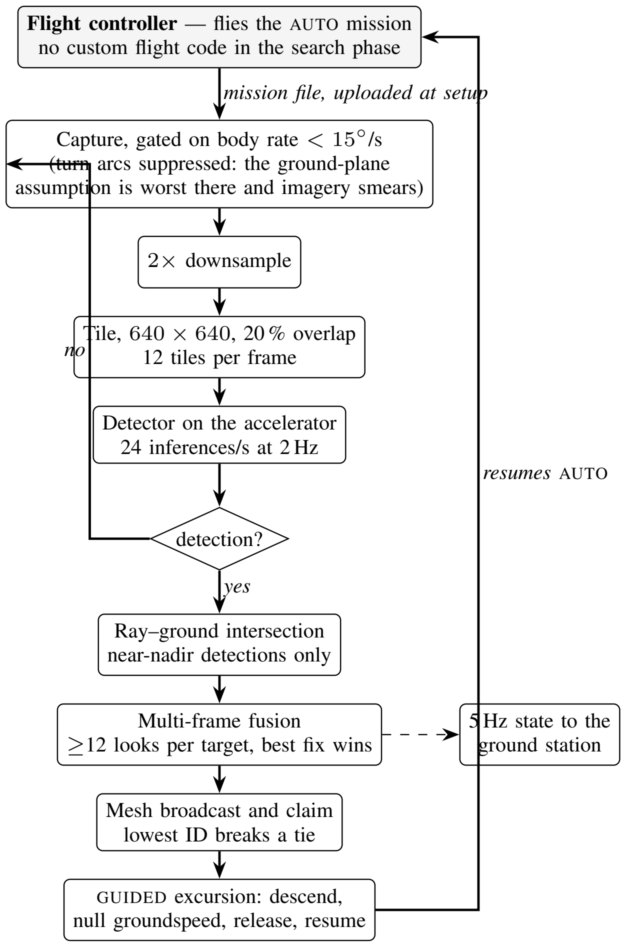

Fig. 4. Onboard processing. The autopilot flies the search pattern unaided; the companion computer detects, geolocates and claims survivors, and interrupts the mission only to deliver. Detection runs alongside the sweep and never gates it — a missed frame costs a look, not a waypoint.

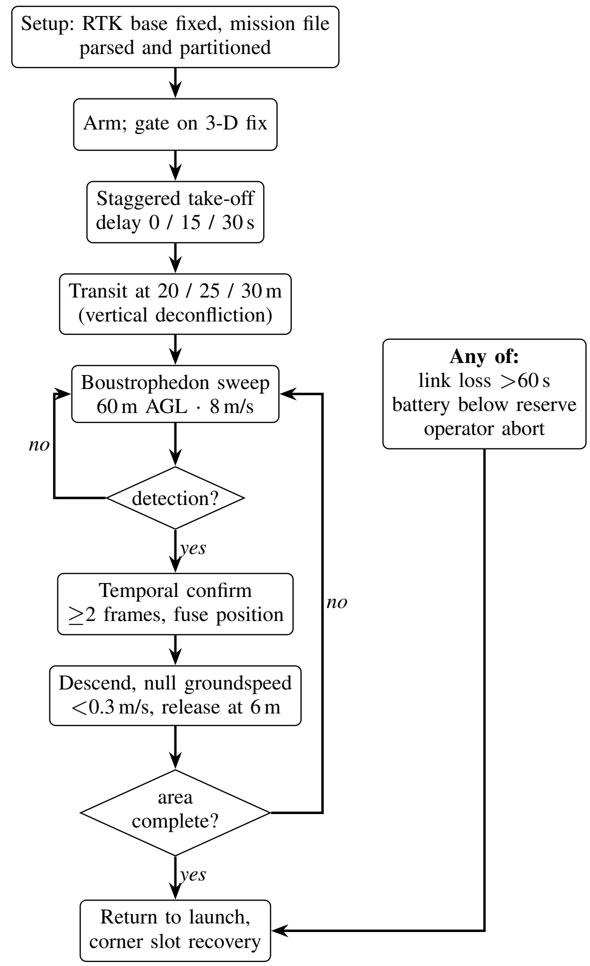

Fig. 5. Autonomous mission sequence. The take-off stagger is encoded in the mission file rather than in operator timing, and all three abnormal conditions converge on a single return-to-launch path.

every terminal branch converges on return-to-launch: link loss, low battery and operator abort share one recovery path rather than three. Third, the battery branch is *staged* rather than single-valued: the reserve threshold commands return-to-launch, and a second, lower threshold commands an immediate landing where the aircraft stands. A pack that cannot reach the pad should land under control at the point it admits this, rather than descend uncommanded partway home.

V. Methodology and Work Plan

The programme is organised into ten phases. Phases P1–P5 are analysis, procurement and integration; P6–P10 are progressively more demanding flight test. The eight-week flight-test window spanned by P6–P10 is the binding schedule constraint on the entire programme, and the work plan is arranged to protect it.

A. Verification strategy

The programme's verification discipline is, I believe, its most transferable contribution, and it exists because of a specific and recurring class of defect.

On four separate occasions during preparatory work, a configuration artefact was found to be correct, unit-tested, reviewed, and *not in the execution path* of the running system. In one instance a validated failsafe parameter set was never loaded by any simulation, so every test ran against stock defaults in which the low-battery action was disabled. In another, a message-routing configuration started cleanly and forwarded zero messages for weeks. Each of these reviewed clean and had passing tests around it.

The countermeasure adopted is procedural rather than technical: no capability is considered demonstrated until the real artefact has been exercised end to end and the resulting values read back off the running system. Consequently every claim in this proposal carries an explicit evidence class, and those classes are maintained mechanically rather than editorially:

- **Measured** — observed on a running system, in software or in hardware-in-the-loop simulation.
- **Modelled** — computed by a model whose inputs are themselves assumptions, and which regenerates in continuous integration.
- **Assumed** — neither measured nor modelled; carried

TABLE III
PROGRAMME PHASES

Ph.	Title	Principal exit criterion
P1	Requirements and sizing	Design point frozen; all models regenerate
P2	Ground segment	Single-operator interface demonstrated against three simulated aircraft
P3	Autonomy	Deterministic plan generation; deconfliction verified in simulation
P4	Fault handling	Link-loss and low-battery behaviour verified by fault injection
P5	Procurement and build	Three airframes assembled and weighed; thrust stand complete
P6	First flight	Hover, trim, vibration and thermal characterisation
P7	Perception field trials	Detection recall measured on real imagery against fabricated targets on RTK-surveyed ground
P8	Delivery trials	Delivery CEP measured against a surveyed target mat
P9	Full mission rehearsal	End-to-end mission with all three aircraft, no operator intervention
P10	Setup and contingency	Setup drill timed over repeated runs; margin recovered

explicitly as a risk.

A continuous-integration job enforces the rule that every published number regenerates byte-for-byte from its model, so that changing a model without committing its regenerated output fails the build. Thirteen model outputs are currently under this gate.

The same discipline now covers the cost basis. Every figure in this section is imported from a single module, which also generates the cost workbook and the budget charts; nothing is restated in two places. That was not true earlier in this programme — the workbook and this document drifted until they disagreed about the headline figure — and the fix was structural rather than editorial, because an inconsistency that can be reintroduced by hand eventually is.

VI. Preliminary Results

Substantial work has been completed before this request. I summarise it here with the evidence class of each result stated, because a claim whose basis is declared can be independently checked, whereas a claim whose basis is left implicit cannot be.

A. Measured in simulation

- **Low-battery failsafe returns the aircraft to the pad.** Verified across three simultaneous software-in-the-loop autopilots; return-to-launch triggered at 10 809 mAh of a 10 800 mAh planned consumption.
- **Three aircraft on one ground station.** A single telemetry router carried three distinct system identifiers to one ground station port with all three arriving.
- **Inter-aircraft separation at launch.** Staggered take-off delays encoded in the mission file, rather than in operator timing, increased the closest airborne approach during launch from 1.31 m to **64.80 m** (Fig. 6). Both figures are recomputed directly from the committed telemetry.
- **Separation during the sweep.** Minimum airborne separation away from the pad is **29.19 m**, with staggered transit altitudes of 20/25/30 m providing vertical deconfliction independent of the lateral plan.
- **Recovery deconfliction.** Assigning landing slots to the corners of the recovery pad rather than in a row increased slot spacing from 1.22 m to 2.61 m at no cost, raising the measured minimum approach from an overlapping 0.83 m to **2.27 m** — a clearance of 1.22 m between airframes (Fig. 7). Separation while stacked in the air over the pad improves by the same mechanism.
- **Ground station.** All eight required operator displays were verified against three simultaneous autopilots, carrying three concurrent H.264 streams within the link budget, with offline map tiles (Fig. 9).

1) Pad containment has exactly zero margin: Fig. 7 shows the corner slots sitting precisely on the inset square. Nominal edge clearance is 0.523 m — one airframe radius — so no part of a stationary aircraft lies outside the box, and the geometric requirement is met by construction.

That is compliance with no margin, and it is worth stating plainly. The rules admit no part of any aircraft outside the box during launch and landing. Touchdown dispersion of ± 0.5 m, the figure this programme uses elsewhere for landing accuracy, would put up to half a metre of airframe over the edge.

The two constraints pull against each other: moving the slots inward to buy edge margin costs inter-aircraft spacing, which is precisely why they were moved to the corners. Resolving it needs a *measured* touchdown dispersion rather than an assumed one, and that is a P6 activity. It is recorded here because the geometry looks safe on the figure and is only safe on average.

B. Modelled

Endurance, hover power, mass budget, thermal margin, link budget, delivery dispersion and geolocation error are all modelled and regenerate from source. The geolocation Monte Carlo reconciles with the independent analytic budget to within one per cent, and the geotag projection has been verified self-consistent between two independent formulations to 7.8×10^{-10} m.

C. Assumed, or not yet implemented

Boresight calibration residual, achieved RTK accuracy in the field, and battery internal resistance under Indian ambient conditions are all assumed, and each is a funded activity in P5–P8. **Detection recall on real post-flood imagery is also assumed, and its field campaign is deferred rather than funded** (Section VII-E); it is therefore the one assumption this

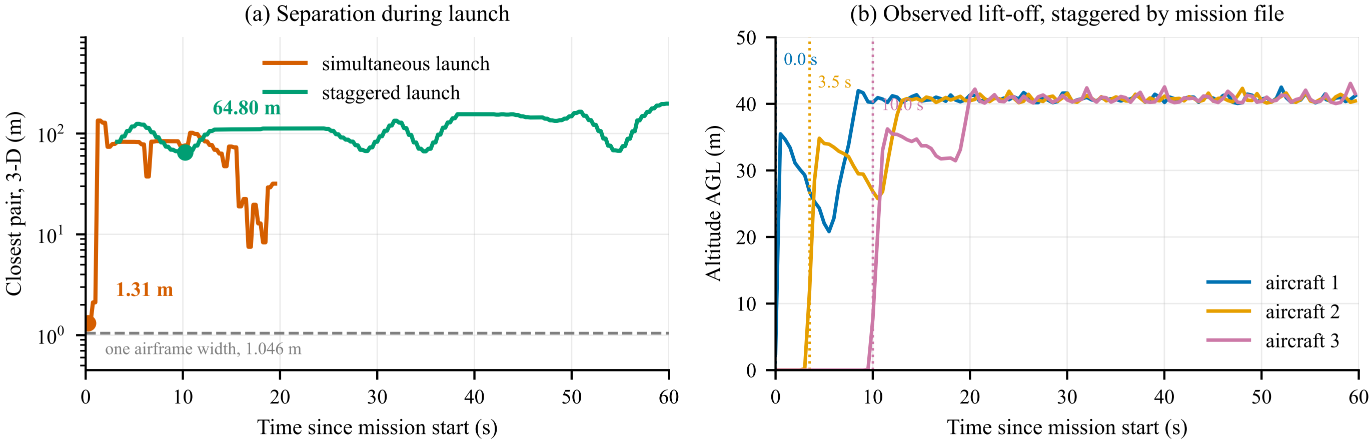

Fig. 6. Launch deconfliction, recomputed from the committed telemetry of three ArduCopter software-in-the-loop autopilots. (a) Closest pair in three dimensions, airborne aircraft only, on a logarithmic scale. Launching simultaneously brings two aircraft within 1.31 m — inside one airframe width, marked — while staggering take-off in the mission file holds the closest pair to 64.80 m. (b) The mechanism: lift-off times observed in the same recording. The mission file commands delays of 0/15/30 s; the aircraft are observed leaving the pad at 0.0/3.5/10.0 s, and **the figure plots what was observed rather than what was commanded** — see Section VI-C.

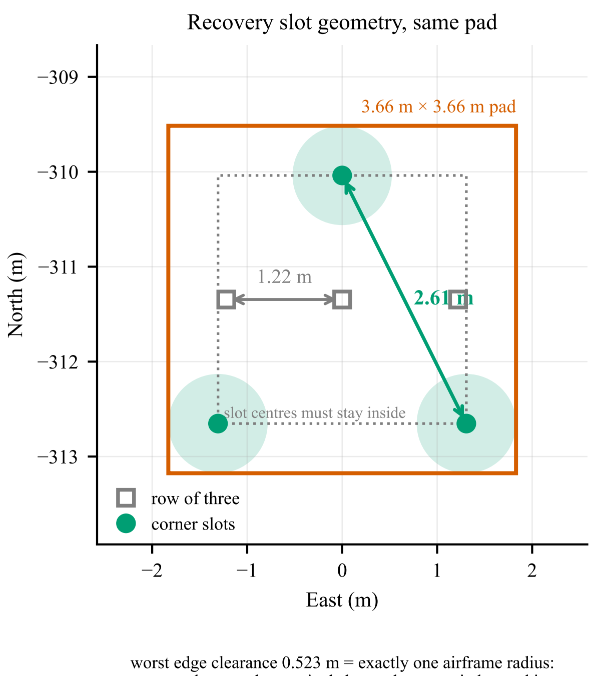

Fig. 7. Recovery slot geometry, drawn from the two committed recordings. Slot centres must stay half an airframe inside the pad edge, so they live in a 2.61 m square: a row across that square gives 1.22 m of spacing, its corners give the full 2.61 m. Shaded discs are the 1.046 m airframe footprint at the corner slots. The measured closest approach at any time rises from an overlapping 0.83 m to 2.27 m — twice the separation, on the same pad, at no cost.

programme does not currently plan to retire, and it leads the risk register accordingly.

One ground-segment function is explicitly *not* implemented and is visible as such in Fig. 9: the abort and recall controls record operator intent in the mission log but transmit nothing, because no downlink command path is yet wired. Recovery today depends on the safety pilot's transmitter. Closing this is a P4 activity, and it now runs over the 2.4 GHz ExpressLRS link rather than the deferred sub-gigahertz radio — which is what makes that link load-bearing for safety and not merely a control convenience.

D. The honest summary

Nothing in this programme has flown. Every "measured" result above was measured in simulation or in software. That rules out whole classes of logical and integration error and rules out none of the physical ones. The requested funding exists principally to convert the modelled and assumed rows of this section into measured ones.

VII. Budget and Cost Analysis

A. Basis of estimate

The cost basis is arithmetic over a bill of materials in which each line carries an exact manufacturer part number, an Indian supplier, and a source. Lines are classified by price provenance: *listed* where a dated Indian retail price was obtained, and *quote required* where no public price exists. I regard the distinction as material, and four of the larger lines — the accelerator, the GNSS receiver, the motors and the cells — still fall in the second category. The autopilot and the camera are dated retail listings.

1) A correction that works against the size of this request: An earlier draft of this budget added 22 % duty and 18 % goods and services tax to the whole subtotal. That was correct for the original bill of materials, whose prices were ex-works budgetary estimates. It became wrong when market verification replaced most of those estimates with *Indian retail listings*, which are duty-paid and tax-paid at the till — the autopilot is quoted by its own manufacturer as "INR 22 600" inclusive, and the deferred Agam board as "INR 42 000 with 5 % GST". **Grossing up a figure that is already gross counts the same tax twice.**

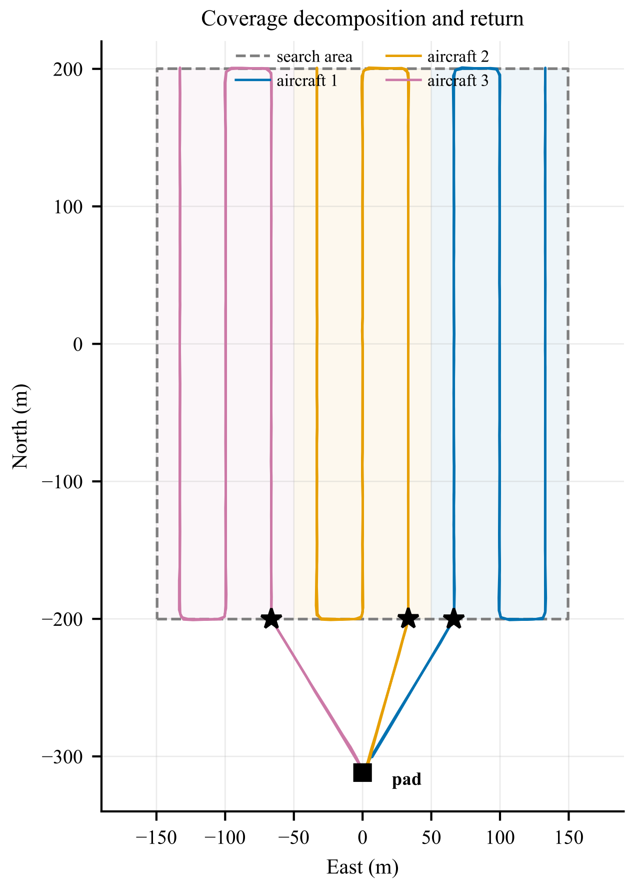

Fig. 8. Coverage decomposition and return geometry, drawn from the committed telemetry. Each aircraft is assigned an equal-area strip (shaded) and flies a boustrophedon pattern within it; each strip is swept twice, on opposing headings, so that boresight error cancels rather than accumulating. Stars mark where each sweep ends. Enumerating the four start-side and direction combinations and keeping the one that finishes nearest home brings all three aircraft to within 117–130 m of the pad, against 516 m and 540 m for two of three under the previous index-keyed rule — at identical path length, and on the lowest state of charge of the flight.

Every line has now been classified, and tax is applied only to what is genuinely untaxed:

- *Tax-inclusive* — a dated Indian retail listing. Nothing is added. INR 4.66 L, the great majority of the bill.
- *Ex-GST* — a supplier quote or a fabrication service, which in India is normally quoted before tax. Duty and GST apply. INR 1.82 L: the GNSS receivers, the cells, the in-house structure and payload fabrication, the RTK base, and the thrust stand.
- *Exempt* — statutory fees. INR 0.05 L.

The correction removes INR 1.64 L of tax that was being charged twice and takes the request from INR 10.12 L to INR 8.24 L. A later pass against the institute's existing equipment removed a further INR 14 500 of ground and test items that turned out to be held on campus. A further review then deferred the remaining spares, the relief kits and the ground-truth apparatus, giving the INR 7.65 L asked for here.

One line then had to move the other way, and it is the one that matters most. Reviewing the compute architecture showed the bill of materials bought a camera and a neural accelerator and no computer to run them. The accelerator is a host-attached module communicating over the single-board computer's PCIe bus, and the camera is a CSI module; neither functions without that host. The fully specified variant carried an edge module with an integrated processor, and the cost reduction replaced it with the accelerator alone without restoring the host. INR 8 000 per aircraft has been added. This is not a change of scope – the aircraft could not have run its detection pipeline as previously costed for here. **Where a line is classified ex-GST on judgement rather than on a document, being wrong makes the request *smaller*, not larger** — the error runs in the sponsor's favour, and confirming the classification with each supplier is a tranche-1 action.

TABLE IV
AIR VEHICLE BY SUBSYSTEM, PER AIRCRAFT: WHAT IS FUNDED, AGAINST THE FULLY SPECIFIED REFERENCE

	Ref.	Adopted		
Subsystem	**INR**	**INR**	**Ind.**	**Driver**
Avionics	85 400	49 100	32 %	Autopilot, RTK
Compute, percep.	79 500	39 100	10 %	Inference rate
Propulsion	57 396	30 600	21 %	Thrust at MTOW
Power	24 500	23 500	60 %	292 Wh reserve
Structure	20 550	13 000	85 %	Arm root moment
Communications	17 800	6 000	30 %	Three feeds
Payload system	5 400	4 500	90 %	Four stations
Total	**2 90 546**	**1 65 800**	**36 %**	

B. The bill of materials in detail

Table IV gives the air vehicle at subsystem granularity. The *Ref.* column is the fully specified aircraft — every line at professional grade, drawn from the verified bill of materials whose 41 entries each carry a manufacturer part number, an Indian supplier and a source link. The *Adopted* columns are what this proposal asks to fund.

Three things in that comparison are worth drawing out.

The order of the subsystems does not change; the reasons do. Avionics is the largest line in both columns and compute the second. What changes is why: in the reference aircraft avionics is expensive *and* comparatively indigenous, because a Pixhawk-standard autopilot is manufactured domestically. In the adopted aircraft that autopilot is imported, and the same subsystem falls to 32 % Indian.

The two largest savings are also the two largest losses of Indian content. Compute falls INR 48 400 and propulsion INR 26 796 — between them 57 % of the total reduction. Propulsion drops from 89 % Indian to 21 %, the sharpest fall in the table, because generic motors and speed controllers replace parts that publish measured thrust.

What the team fabricates itself becomes more Indian, not less. Structure rises from 79 % to 85 % and the payload system

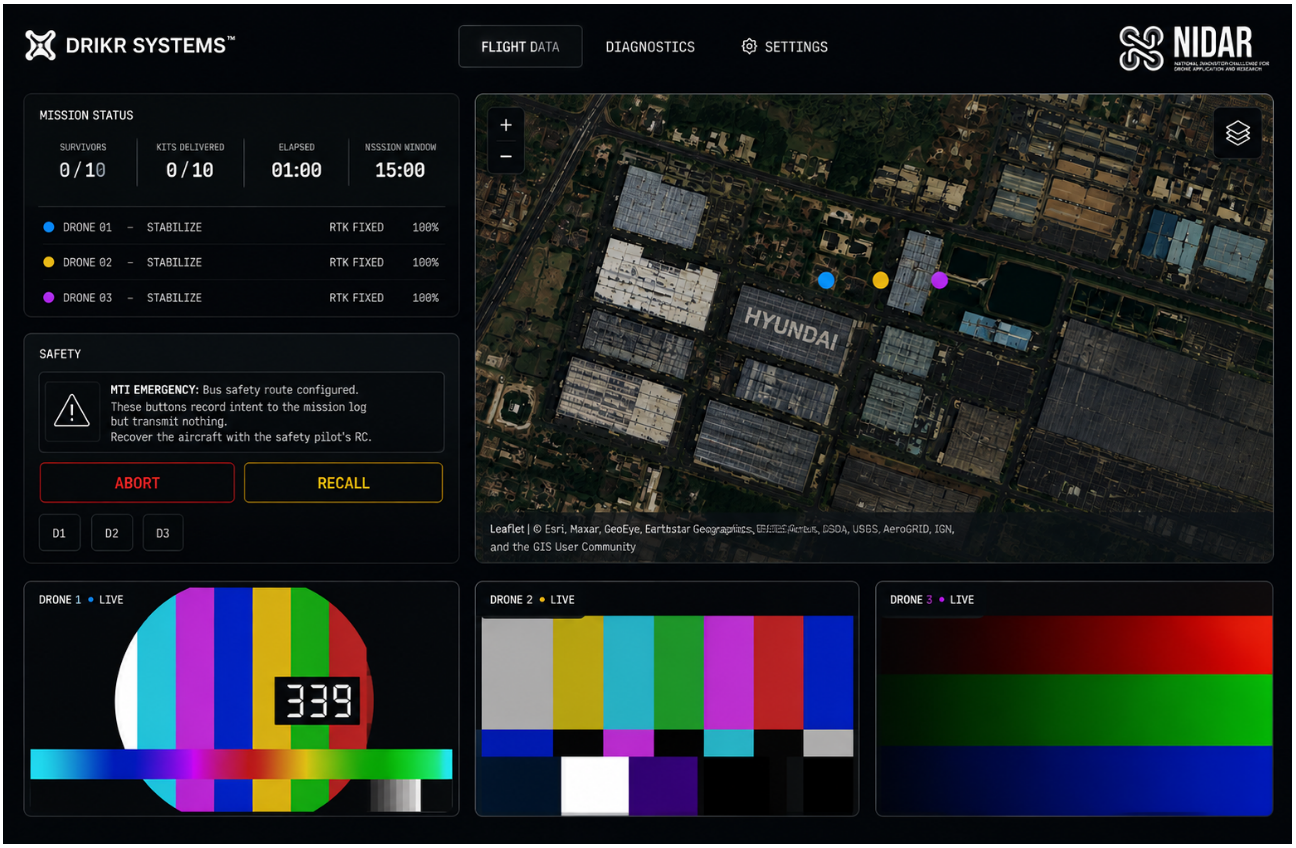

Fig. 9. Ground control station, captured from the running system. A single operator interface carries all eight displays the rules require: mission status, per-aircraft mode, RTK fix state and battery, the shared map, and three concurrent H.264 feeds within the 2.5 Mbps link budget. Map tiles are served from a local cache, so nothing here depends on a network. The safety panel is shown *uncropped*: abort and recall record operator intent in the mission log and transmit nothing, because no downlink command path is yet wired — see Section VI-C. The video panes carry test patterns; no camera has yet flown.

from 61 % to 90 %, because in-house machining and printing displace bought assemblies. Power holds at 60 %: the cells stay Indian and BIS-certified, and that was a deliberate retention rather than a residue of the cost pass.

Compute remains the least indigenous subsystem in both columns, for the single reason developed in Section VIII: no Indian accelerator exists in this class.

C. Cost reduction study

A structured cost-reduction study was performed against the verified bill of materials, producing four costed configurations of the same aircraft plus one reduced-capability alternative. Fig. 10 summarises them, and Table IV shows, subsystem by subsystem, where the adopted configuration takes its saving.

The options are ordered by *what they trade*, not by descending price, because the interesting question is not how cheap the aircraft can be made but what each rupee costs in capability or sovereignty.

Option D is adopted, at INR 1 57 800 per aircraft. The governing principle is not "buy cheaper" but *hobby-grade where the failure mode is visible and the specification is easy to verify; professional where it is not.*

Motors, electronic speed controllers, propellers, adapters, pack hardware, the mesh node and mounts are commodity parts whose ratings can be checked on a bench before they fly. Six lines are held at professional grade on purpose, and the reason is recorded against each: the autopilot (dual IMU, Pixhawk standard), the RTK receiver (no cheaper part exists, and it governs 125 of 200 geolocation points), the accelerator (26 TOPS is the specification floor rather than a preference), the camera (ground sample distance drives detection), the cells (capacity- and IR-matched, because one weak cell defines the whole pack), and the airframe structure.

This is bought with margin, not with capability. Every scored function survives: RTK geolocation, onboard detection at rate, three concurrent video feeds, four-station delivery. What is given up is reserve — published thrust data, supplier support, spares depth and indigenous content.

The accelerator substitution carries an integration consequence I did not see at first and record here because it enlarges the benchmark rather than the bill. The displaced module has a *hardware* H.264 encoder; the single-board computer that replaces it has none — its GPU dropped every encode block, so all video encoding is done in software on the same CPU that feeds the accelerator. Each aircraft encodes only its own stream, so this is one 720p encode per board and is achievable, but it is not free. **The P5 benchmark is therefore not merely whether the accelerator reaches 24 inferences per second;**

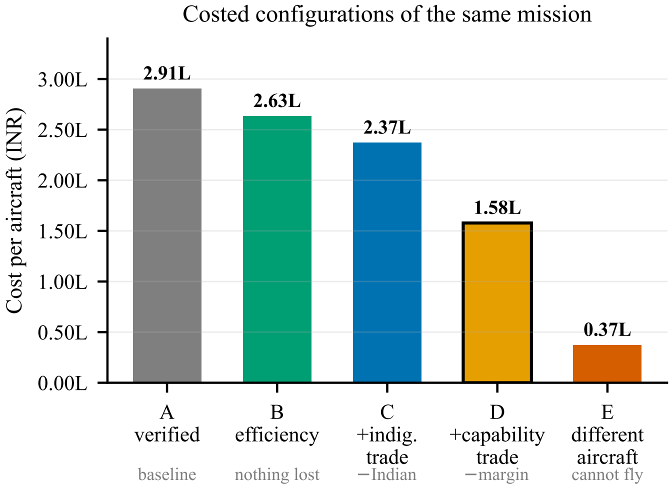

Fig. 10. Costed configurations, ordered by what each trades away. A through D are the same aircraft; E is a different and much smaller airframe that cannot carry the payload, included to show where the cost floor genuinely lies rather than as a proposal. Option D is adopted: it gives up margin and indigenous content, but no scored capability.

it is whether the board sustains that rate and a 720p H.264 encode simultaneously without thermal throttling. The cooling line has been left untouched for the same reason.

The generic motors move an existing risk rather than creating one. The selected parts carry no published thrust curve, where the displaced ones did. That does not change the physics — it changes who measures it, and when. The thrust stand, carried at INR 8 000 in the test-equipment line, is therefore no longer a convenience but the instrument that decides whether the aircraft hovers at the design point.

It cannot precede procurement, because it needs the motors it is characterising. Thrust is therefore measured in P5 as the motors arrive, and **a shortfall against the datasheet at that point forces a propulsion change with the flight-test window already open.** That is the honest shape of the risk, it is why the entry in Table IX is rated High, and it is a direct cost of choosing parts that publish no measured thrust.

1) Corrections made to this analysis on review: A first pass at this study reported a floor of INR 1 29 632 and attributed a 125-point geolocation penalty to Option C. Both were wrong, and the errors are recorded here rather than quietly fixed, because a reviewer who cannot see the corrections cannot judge the method.

Option C does not lose RTK, and its geolocation penalty is negligible. The saving in question deletes the second GNSS receiver, which supplies moving-baseline heading. Substituting the magnetometer-class attitude allocation of Table II into the case-C budget moves the total from 1.304 m to 1.322 m — an 18 mm degradation, not a scoring event. The 125-point penalty belongs to Option D, which deletes RTK itself.

Three savings were withdrawn as unsound. A motor substitution booked INR 8 396 at a price below the lowest listing found for any comparable part, which is circular given that the same study argues the carried price is optimistic. An ESC substitution booked INR 1 600 against a part whose catalogue advertises 30–45 A for a 60 A requirement, which is not a substitution but a specification failure. A sub-gigahertz radio was priced at INR 1 560 in the lowest option while the same document argued that figure does not buy a compliant 865–867 MHz link. Compliance is not an option-level choice, so the compliant price is now carried in every column.

The floor moved upward by INR 54 008 as a result — from INR 1 29 632 to INR 1 83 640. The conclusion this strengthens rather than weakens: an aircraft that flies this mission cannot be built for INR 1 00 000, and the corrected figure is further from that target than the first estimate was.

2) Why the second GNSS receiver is deferred to a measurement: Given that its budgeted contribution is 18 mm, the honest case for the second receiver was never the error budget — it is insurance against a risk the budget does not model. The budget allocates a small attitude term on the assumption of good heading. A magnetometer mounted near a 98 A power bus may not deliver that: at the frame edge, 37 m off nadir, a heading error of 5° displaces a detection by 3.22 m, which would consume the entire delivery allowance on its own.

That is a real risk, but it is an *unmeasured* one, and INR 51 000 is a great deal to pay to avoid a measurement the programme will make regardless. **P6 characterises achieved heading accuracy on the built aircraft at no additional cost.** If it exceeds roughly 1° the receiver is reinstated and the money is well spent; if it does not, it was never needed. Deferring on evidence rather than on funding is the distinction the reinstatement table in Section VII-E draws, and this is the line that motivated it.

Options A to C are retained in Fig. 10 as the reasoning that led to D, not as live alternatives: each is a step in the argument, and a reviewer should be able to see the sequence rather than only its conclusion. Option E is a reduced training platform that cannot carry the payload at all and is discussed in Section VII-G.

D. What is actually being requested

An earlier version of this budget costed a development *programme* at INR 28.74 L. That figure bought laboratory equipment, computing and a purchased field-data campaign. On review, none of those are what a competition entry needs, and most of the equipment is already held by the host institution. The ask has been recosted accordingly and is now **INR 7.65 L**, a reduction of 71 %. Table V gives the breakdown.

E. What is deferred, and what it would cost to reinstate

The ask above is the minimum that flies the scored mission. Several items were deliberately deferred rather than judged unnecessary, and Table VI lists them in the order I would reinstate them against additional funding. Stating them is not padding: a reviewer should be able to see what the lowest-cost configuration gives up.

Ground-truth apparatus is fabricated rather than bought, taking that line from INR 35 280 to INR 4 860. Survivor targets

TABLE V
COMPETITION BUILD COST

Element	Was	Ask
Air vehicles (3) and payload	8.31 L	4.97 L
Ground segment	3.88 L	0.56 L
Test equipment	2.26 L	0.34 L
Spares and consumables	2.73 L	0.13 L
Regulatory	1.75 L	0.05 L
Subtotal	18.93 L	6.06 L
Duty and freight, on ex-tax lines only		0.24 L
GST, on ex-tax lines only		0.35 L
Contingency, 15 %		1.00 L
Total requested		**7.65 L**

Rows are rounded to 0.01 L and may not sum exactly; the total is computed from unrounded figures.

TABLE VI
DEFERRED ITEMS, IN REINSTATEMENT PRIORITY

#	Item	INR	What it buys back
1	Flight insurance	25 000	**Must be confirmed** — see below
2	Second GNSS receiver (3)	51 000	**Conditional on a measurement**, not on funding
3	Recovery parachutes (3)	36 000	∼40 points per crash avoided
4	Training airframe	37 309	Protects the fleet in week 1
5	Sub-GHz safety radio (3)	58 065	A third independent link
6	Spare packs and motors	74 396	Survives a crash mid-campaign
7	Field data campaign	15 000	Detection recall measured, not modelled
8	Spare compute module	20 000	Removes a single point of failure
	Total deferred	**3.74 L**	

are clothed forms of the correct size and outline; at 60 m and 2 cm/px a person subtends about fifty pixels, and what the detector responds to is silhouette, scale and contrast against terrain, not anatomy. Two of these items are *better* fabricated: dummy kits can be made to exactly the 200 g that rule C6 fixes, and donated clothing gives wider colour and contrast variety than matched bought sets, which is precisely what detection robustness requires.

One caveat follows from this and is recorded rather than glossed. The calibration markers are surveyed using the programme's own RTK rover. That is standard practice and adequate for this purpose, but it means the ground truth against which geolocation accuracy is measured is *itself RTK-derived*. The two error sources are therefore correlated, and the measured geotag figure reported in P7–P8 will be slightly optimistic against an independently surveyed datum. Any accuracy claim arising from those phases will state this.

Four further items deserve comment rather than a table row.

The second GNSS receiver is deferred to a measurement, not to funding. It supplies moving-baseline heading, and Section IV-C shows that substituting a magnetometer-class attitude allocation into the case-C budget costs **18 mm**. The honest case for it was never the error budget — it was insurance against a magnetometer sitting near a 98 A bus, which the budget does not model. That is an unmeasured risk, and INR 51 000 is a great deal to pay to avoid a measurement I intend to make anyway. **P6 characterises achieved heading accuracy on the built aircraft; if it exceeds roughly 1°, the receiver is reinstated** and the money is well spent. If it does not, it was never needed.

The sub-gigahertz safety link is deferred on an architectural argument. The ExpressLRS link at 2.4 GHz carries safety-pilot control and abort independently of the autonomy stack, and the mesh at 5.8 GHz carries video, telemetry and coordination. Those are already two independent bands across a 600 m geofence, and the ExpressLRS link covers that range with substantial margin. The 865–867 MHz set is a *third* path. It is the correct thing to add when funding allows, and it is not the thing to buy before the aircraft flies.

That argument holds only for one specific radio configuration, so I state it as a build requirement rather than leave it implicit. The receivers must run ExpressLRS 3.5 or later in *native* MAVLink mode, which interleaves RC control and MAVLink on a single link and a single autopilot UART [9]. The older AirPort mode reaches the same autopilot by a transparent serial bridge, but it *consumes* the link: a build configured that way has telemetry and no control, and needs a second radio to fly — which is precisely the sub-gigahertz set deferred here. Configured wrongly, the deferral is not a saving but a defect, and the failure appears at integration rather than at review.

The abort path inherits the availability of that link. Because abort and recall ride the ExpressLRS link, an outage removes the operator's fastest intervention until the link returns or the 60 s link-loss timer fires the autonomous return of Fig. 5. The aircraft is never without a recovery path — that is the point of routing every terminal branch to return-to-launch — but it can be briefly without an *operator-commanded* one. Measuring outage statistics against range and attitude is a P6 objective, and it is the evidence that decides whether the sub-gigahertz path is a nice-to-have or a requirement.

Insurance is deferred at the team's direction and I flag it as the one item that may not be optional. Third-party cover is commonly mandatory for unmanned operations in India. It must be confirmed against the rulebook and the prevailing DGCA rules **before any flight**, and if it is required it ceases to be a budget decision.

The recovery parachute is permitted, not required. An earlier draft of this proposal stated that a requirement mandated it; that was wrong. The organisers permit ballistic recovery, and the case for fitting one is a points argument rather than a compliance one — a crash is scored at -50 against -10 for landing outside the recovery zone, so deployment is worth roughly 40 points before the value of the airframe is counted. Deferring it is an accepted risk, taken knowingly.

No spectrum licence is required and none is budgeted. Every link in the system operates in a delicensed band: 2.4 GHz and 5.8 GHz ISM for the mesh and control links, and 865–867 MHz for the independent safety link. Equipment type approval is a supplier obligation, discharged by purchasing approved radios from Indian vendors, so only DGCA aircraft registration remains as a regulatory line.

Four principles were applied, and each is visible in the table.

If the institution already owns it, it is not in the ask. **INR 3.54 L of equipment** is available and has been removed: the electronics laboratory and 3D printing (INR 84 000), GPU compute for model training (INR 75 000), field power (INR 55 000), and the machine shop and composite fabrication that produce the CNC-cut plates, machined arm clamps and motor mounts. Both ground-station laptops (INR 1,40,000) are supplied by the team. None of this is deleted — it is recorded as an **institutional contribution**, because that is what it is: co-funding in kind, and the reason the ground segment falls from INR 3.88 L to INR 1.37 L.

The remaining ground-segment spend is dominated by one item that could *not* be borrowed: the RTK base receiver and its survey tripod (INR 47 500). Section IV-C explains why it is not a candidate for deletion — it governs the entire geolocation budget.

If it can be built, it is costed as built. The thrust stand falls from INR 45 000 to INR 8 000 — a load cell, an amplifier and a printed fixture. Motor thrust verification is a flagged risk in Table IX; building and calibrating the instrument is a better answer to that risk than a purchase order for one.

Spares cover what crashes, not what fails rarely. Propellers, motors, arms, plate stock and packs are retained. Spare autopilot, GNSS and camera — INR 60 600 together — are removed. **This is an accepted risk and I state it as one:** a failure of any of those three during the flight-test window costs schedule, and the mitigation is procurement lead time rather than inventory.

What is deferred is listed, not hidden. Section VII-E sets out every item removed to reach this figure, in the order I would reinstate it against additional funding. The lowest-cost configuration is a deliberate choice, and a reviewer should be able to see exactly what it gives up.

The contingency is held at 15 % rather than 20 %, which the phased disbursement in Section VII-F makes defensible: unreleased tranches are themselves a reserve, and holding both is asking for the same margin twice.

F. Phased disbursement

The full sum is not required at once. The programme is structured as four tranches tied to the phase exit criteria of Table III, and I would rather be funded against demonstrated progress than hold a sponsor's capital for eight months. Table VII and Fig. 11 give the schedule.

Three properties of this schedule are deliberate.

TABLE VII
PHASED DISBURSEMENT SCHEDULE

T	Months / phases		INR	Release condition
1	1–2	P1–P4	2.77 L	On award. Design point frozen; models regenerate
2	3–4	P5	2.24 L	P1–P4 complete: ground segment demonstrated, and link-loss and low-battery behaviour verified by fault injection
3	5–6	P6–P8	1.60 L	Three airframes weighed; first hover complete
4	7–8	P9–P10	1.04 L	Perception and delivery trials reported with measured figures
	Total		**7.65 L**	

Tranche 1 is front-loaded and mostly procurement. The largest single schedule risk in the programme is that long-lead components arrive late and compress the flight-test window. Cells, autopilots and GNSS receivers are ordered in month one, before most of the software work is finished, because the lead time is the constraint rather than the readiness of the integration.

Each tranche is released against a stated, checkable exit criterion drawn from Table III — not against a date and not against a progress report. A sponsor can verify every one of them from the artefacts the programme already produces.

Tranche 4 is the smallest and holds the contingency. If the programme underperforms, the unreleased balance is the sponsor's, and the exposure at any point is bounded by the cumulative curve in Fig. 11 rather than by the total.

G. A note on the training platform

Option E in Fig. 10 is a INR 37 309 training airframe. It fails eight system requirements and cannot fly the mission; it is emphatically not a low-cost variant of the proposed system. Its value is different: the programme's binding constraint is an eight-week flight-test window currently budgeted entirely on aircraft that do not yet exist and cannot be risked once they do. A training airframe, at roughly twice the cost of one inference module, would fly from the first week of that window and absorb ground-station integration, radio-path validation, setup-drill timing, failsafe verification on real hardware, and crash-and-rebuild practice.

extbfIt is deferred in the present ask and sits at position 4 in Table eftab:deferred. I record it as the highest return-on-investment discretionary item in the budget rather than quietly dropping it: with the spares set also reduced, it is the mitigation the flight-test window most obviously lacks.

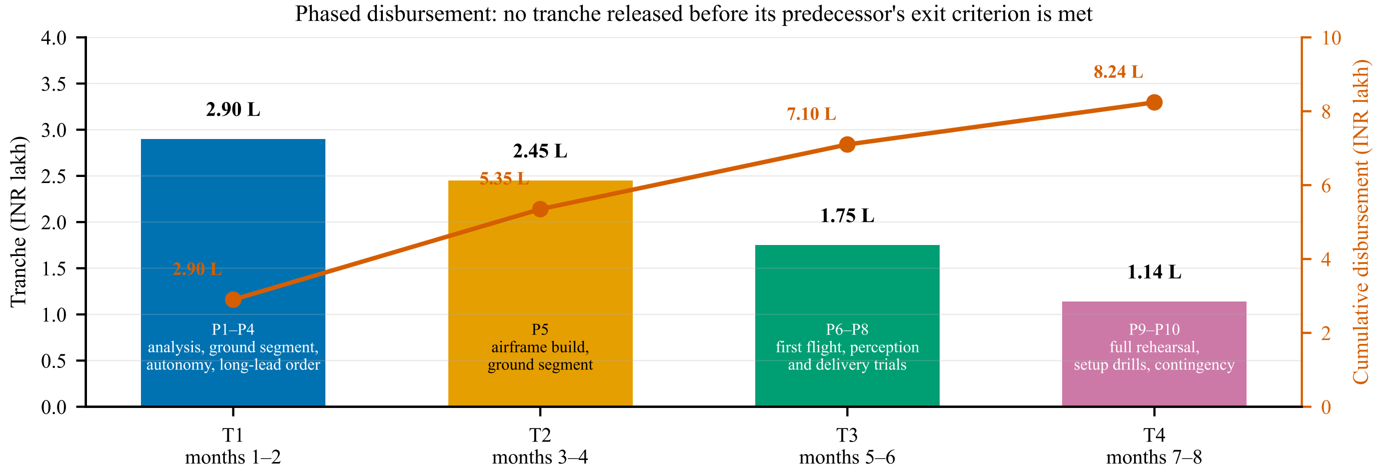

Fig. 11. Phased disbursement against programme phases. Tranche 1 is front-loaded because long-lead components must be ordered in month one for the eight-week flight-test window to survive; tranche 4 is the smallest and carries the contingency.

VIII. Indigenisation Strategy

Indigenous content is measured, not asserted. Every line in the bill of materials carries a declared Indian value fraction, and the aggregate is computed rather than estimated, which makes the claim auditable.

The configuration adopted in Section VII-D carries approximately 36 % Indian content, down from 58 % in the fully specified aircraft. That figure is computed from the per-line Indian fractions of the adopted bill of materials rather than estimated; an earlier draft of this proposal asserted 45 %, which the computation does not support. I report what applies to what is actually being funded rather than the figure that flatters the argument.

Fig. 12 decomposes it, and the decomposition is far more informative than the aggregate. **Propulsion falls from 89 % to 21 % and avionics from 60 % to 32 %** — the two lines that make the aircraft affordable are precisely the two that make it less Indian. What remains strongly domestic is what the team fabricates or assembles itself: the payload system at 90 % and the airframe structure at 85 %.

This is the clearest cost of the budget decision, and it is reversible. Two lines account for almost all of it: restoring the Indian autopilot costs INR 19 400 per aircraft and restoring Indian propulsion — motors, speed controllers and certified propellers — costs approximately INR 26 800 per aircraft. Together that is INR 1.39 L across the fleet, and it would return the aircraft to roughly 58 % Indian content. Both are drop-in reinstatements requiring no redesign. Component sponsorship from the Indian suppliers surveyed for this bill of materials is the intended route, and it is the reason those suppliers are named throughout rather than described generically.

The inference accelerator is the one genuinely irreducible import, and I declare it plainly rather than obscuring it. Mitigation is confined to what is honest: an Indian carrier board, and Indian enclosure and thermal design. I regard an explicit gap analysis as more credible than a uniformly confident claim,

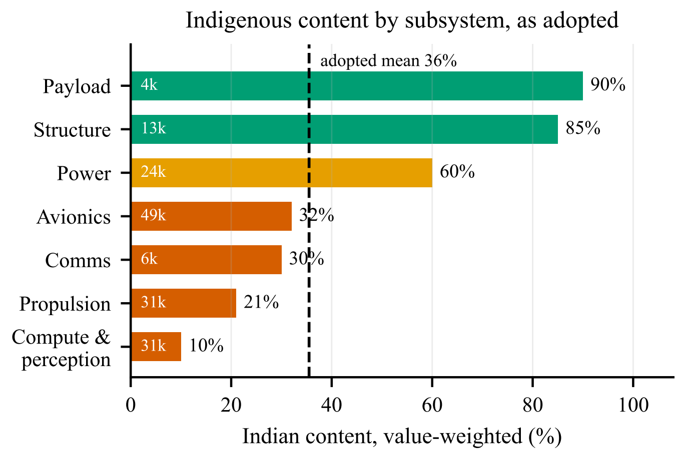

Fig. 12. Indian content by subsystem for the adopted configuration, value-weighted, with subsystem cost annotated inside each bar. Four subsystems now sit below 40 %, and between them they carry INR 1.17 L of the INR 1.58 L aircraft — which is why the aggregate is worth decomposing rather than quoting alone. What remains strongly domestic is what the team fabricates itself.

and I note that the domestic RISC-V development kit carried on the aircraft is a genuine demonstration of Indian silicon in flight, albeit in a monitoring role rather than as an accelerator.

IX. Risk Register

I draw particular attention to the setup-time risk. It is currently supported by a model with a modest margin and no measurement, and it gates the entire mission. Timed bench drills are scheduled as the earliest hardware activity in the programme for precisely this reason.

X. Expected Outcomes and Deliverables

1) Three flight-qualified aircraft with a documented, market-verified bill of materials at approximately 36 % Indian content, and a costed route to restoring the deferred Indian autopilot and propulsion.

TABLE VIII
INDIGENISATION STATUS BY SUBSYSTEM

Subsystem	Status
Autopilot	*Deferred:* Indian Pixhawk-standard board, domestically made
Propulsion	*Deferred:* Indian motors and ESCs with published thrust data
Airframe	Indian composite fabrication
Cells	Indian, BIS-certified 21700 NMC
Camera	*Deferred:* Indian embedded-vision supplier
GNSS	Indian receiver, NavIC L1+L5. *Retained*
Co-processor	Indian RISC-V development kit
Inference module	**No Indian equivalent exists at this class**

2) A single-operator ground control station demonstrated end to end against real aircraft, with no network dependency.
3) A measured detection recall figure against fabricated survivor targets on surveyed ground, replacing the current assumption. Ground truth is RTK-derived rather than independently surveyed, and the resulting figure will state that.
4) A measured delivery circular error probable against a surveyed target, replacing the current model.
5) A measured setup-to-launch time distribution over repeated drills.
6) An open, reproducible evidence base in which every published figure regenerates from its model under continuous integration.
7) A supply-chain report identifying, for each subsystem, the domestic capability that exists, the capability that does not, and what would be required to close the gap.

The seventh deliverable may prove the most durable. The exercise of sourcing a complete airframe domestically, at component granularity and with verified prices, produced a map of Indian UAV supply-chain maturity that I have not seen published elsewhere, including the specific and narrow point at which it currently fails.

XI. Future Work

This proposal deliberately asks for a competition build and nothing more. The work has a longer arc, and I set it out here so that the scope of the present request is unambiguous by contrast.

A. From a competition entry to a deployable system

The entry is not a product. Reaching one requires, in rough order: night and all-weather capability through thermal imaging, which is the single highest-value sensing extension and is excluded here only because it doubles the perception budget; beyond-visual-line-of-sight approval, which is a regulatory programme rather than an engineering one; ingress protection and thermal qualification for monsoon conditions; and endurance beyond the present 15.3 minutes, which at this payload means a different airframe class rather than a better battery.

B. Cost at production volume

Every figure in Section VII is single-unit retail for a build of three. The marginal cost of an additional aircraft is already substantially below the per-aircraft figure quoted here, because the ground segment, test equipment and spares are one-off. At production volume the relevant questions are different ones — tooling amortisation, negotiated component pricing, and assembly labour — and none of them can be answered honestly from a three-aircraft bill of materials. I decline to extrapolate them.

C. A route to commercialisation

The natural customers are state disaster management authorities and the national and state disaster response forces. Establishing whether that is a viable business requires three things this programme does not attempt: the size and refresh rate of the addressable fleet, what a comparable capability is currently procured for, and the operating cost per flight hour of the boat or helicopter search it would displace. Those are questions for published tender data and operator interviews, not for a bill of materials, and I would rather name them as open than fill them with plausible figures.

What this programme does produce toward that end is the seventh deliverable in Section X: a component-level map of where Indian UAV supply chain capability exists and where it does not. That artefact outlives the competition regardless of what follows it.

XII. Conclusion

I propose an autonomous three-aircraft system that addresses the localisation bottleneck in post-flood search and rescue, sourced domestically wherever the requested budget permits, and engineered under a verification discipline that distinguishes what has been demonstrated from what has merely been designed. Where an Indian part was deferred it was deferred on cost, not on capability — domestic industry can supply the autopilot and the propulsion, and Section ef-sec:indig states what restoring them costs.

The system's design point, error budgets and cost basis are established and reproducible. Its coordination and failsafe behaviour have been demonstrated in hardware-in-the-loop simulation across three simultaneous autopilots. What remains is the part that requires funding: to build the aircraft, fly them, and replace the modelled and assumed quantities in Section V-A with measured ones.

I am asking for INR 7.65 L to do that, and I have tried throughout this document to make every number in it checkable.

Acknowledgment

I thank the Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, for institutional support.

References

[1] A. C. Kapoutsis, S. A. Chatzichristofis, and E. B. Kosmatopoulos, "DARP: Divide areas algorithm for optimal multi-robot coverage path planning," *Journal of Intelligent & Robotic Systems*, vol. 86, pp. 663–680, 2017.

[2] H. Choset, "Coverage of known spaces: the boustrophedon cellular decomposition," *Autonomous Robots*, vol. 9, no. 3, pp. 247–253, 2000.

[3] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 912–926, 2009.

[4] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *Proc. IEEE International Conference on Image Processing (ICIP)*, 2022, pp. 966–970.

[5] S. Sambolek and M. Ivasic-Kos, "Automatic person detection in search and rescue operations using deep CNN detectors," *IEEE Access*, vol. 9, pp. 37 905–37 922, 2021.

[6] P. Zhu *et al.*, "Detection and tracking meet drones challenge," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7380–7399, 2021.

[7] S. G. Mathisen, F. S. Leira, H. H. Helgesen, K. Gryte, and T. A. Johansen, "Autonomous ballistic airdrop of objects from a small fixed-wing unmanned aerial vehicle," *Autonomous Robots*, vol. 44, pp. 859–875, 2020.

[8] Internal Displacement Monitoring Centre, *Global Report on Internal Displacement 2025*, Geneva, 2025. [Online]. Available: https://www.internal-displacement.org/countries/india/

[9] ExpressLRS Project, "MAVLink," ExpressLRS documentation, 2026. [Online]. Available: https://www.expresslrs.org/software/mavlink/

[10] ArduPilot Development Team, "ArduPilot: open source autopilot software suite," 2026. [Online]. Available: https://ardupilot.org/

[11] T. Takasu and A. Yasuda, "Development of the low-cost RTK-GPS receiver with an open source program package RTKLIB," in *Proc. International Symposium on GPS/GNSS*, 2009.

TABLE IX
PRINCIPAL RISKS AND MITIGATIONS

Risk	Sev.	Mitigation
Detection recall below target on real imagery	High	**Raised by deferral of the field data campaign.** Public aerial-person datasets carry the model; tiling and fusion parameters remain tunable without hardware change
Inference and software video encode contend on one board	High	The replacement compute has no hardware H.264 encoder. Benchmark both loads together before design freeze; fallback is a reduced tile count on a centre crop
RTK capability unconfirmed on selected receiver	High	Supplier confirmation is a gating procurement action; degraded mode already specified and flagged to the operator
Flight-test window compressed by procurement delay	High	**Raised by deferral of the training airframe and of spares.** Lead time is now the only mitigation, so long-lead items are ordered in tranche 1
Recovery parachute sizing and supply	Medium	No domestic supplier identified; import lead time and correct mass rating are open procurement actions
Setup time exceeds allowance	Medium	Currently modelled, not measured; timed drills are the highest-priority early test
Airframe overhangs the pad edge on touchdown	Medium	Nominal clearance is exactly one airframe radius, so dispersion has no margin. Measure touchdown dispersion in P6 before accepting the slot geometry
Motor thrust below the datasheet claim	High	Selected motors publish no thrust curve, and the stand cannot precede the motors. Thrust is measured in P5 on arrival; a shortfall then forces a propulsion change with the flight-test window already open
Airframe loss during test	High	**Raised by deferral of spares.** Structure set and consumables retained; a second loss in the same week is uncovered
Flight insurance may be mandatory	Medium	Deferred at team direction. To be confirmed against the rulebook and DGCA rules before any flight
Tax classification of quoted lines unconfirmed	Low	The per-line audit is done and INR 1.64 L of double-counting removed. Remaining risk is only that a line marked ex-GST is in fact inclusive, which would reduce the request further